



Laboratory Schedule:

Session No	Topics: Programming Assignments	No. of hours
1.	Creating and Running Simple C Programs: • C-Program to calculate the sum of three numbers / C-Program to demonstrate a Simple Calculator. • C-Program to calculate the area and circumference of a circle using PI as a defined constant. • C-Program to convert temperature given in Celsius to Fahrenheit and Fahrenheit to Celsius • C-Program to compute the roots of a quadratic equation by accepting the coefficients.	2hr
2.	Creating and Running C Programs on Expressions : • C-Program to calculate quotient and remainder of two numbers.) • C-Program to evaluate two complex expressions. • C-Program to demonstrate automatic and type casting of numeric types • C-Program to calculate the total sales given the unit price, quantity, discount and tax rate • C-Program to calculate a student's average score for a course with 4 quizzes, 2 midterms and a final. The quizzes are weighted 30%, the midterms 40% and the final 30%.	2hr
3.	Creating and Running C Programs on Making Decision: • C-Program to determine the use of the character classification functions found in c-type library. • C-Program to read a test score, calculate the grade for the score and print the grade. • C-Program to uses a menu to allow the user to add, multiply, subtract and divide two numbers using switch case. • C-Program to read the name of the user, number of units consumed and print out the charges. An electricity board charges the following rates for the use of electricity: • For the first 200 units 80 paise per unit • For the next 100 units 90 paise per unit • Beyond 300 units Rs 1 per unit. All users are charged a minimum of Rs. 100 as meter charge. If the total amount is more than Rs 400, then an additional surcharge of 15% of total amount is charged	2hr
4.	Creating and Running C Programs on Repetition or Loops: • C-Program to print a number series from 1 to a user-specified limit in the form of a angle	2hr

	• C-Program to print the number and sum of digits in an integer. • C-Program to calculate the factorial of a number using for loop/ Recursion • C-Program to calculate nth Fibonacci number. • C-Program to convert binary to a decimal number	
5.	Creating and Running C Programs on One Dimensional Arrays: • C-Program to print square of index and print it. • C-Program to calculate average of the number in an array. • C-Program to sort the list using bubble sort. • C-Program to search an ordered list using binary search.	2hr
6.	Creating and Running C Programs on Two Dimensional Arrays: • C-Program to perform addition of two matrices. • C-Program to perform multiplication of two matrices. • C-Program to find transpose of the given matrices. • C-Program to find row sum and column sum and sum of all elements in a matrix. • C-Program initialize/fill all the diagonal elements of a matrix with zero and print	2hr
7.	Creating and Running C Programs on User Defined Functions: • C-program to read a number, Find its factorial using function with argument and with return type. • C-Program to read two number, Find its GCD and LCM using function with arguments and without return type. • C-Program to read a number, Find whether it is a palindrome or not using function without argument and with return type. • C-Program to read a number, Find whether it is prime number or not using function without arguments and without return type.	2hr
8.	Creating and Running C Programs on Strings: • C-program read two strings, Combine them without using string built-in functions. • C-program read two strings, Compare them without using string built-in functions. • C-program read two strings, concatenate them without using string built-in functions. • C Program to Check if the Substring is Present in the Given String. • C-program to demonstrate built-in sting functions like strlen(), strcpy(), strcmp(), strcat()	2hr

9	Creating and Running C Programs on Storage Classes and Pointers: • C-program to show the use of auto and static variable. • C-program to add two numbers using pointers. / C-program to swap two numbers using pointers. • C-program to show how the same pointer can point to different data variable. / C-program to show the use of different pointers point to the same variable • C-Program to read an array of elements, Compute its sum using pointers.	2hr
10	Creating and Running C Programs on Derived Types and Unions: • C-Program to print selected TV stations for our cable TV systems. • C-Program to demonstrate union of short int and two char	
11	Creating and Running C Programs on Structures: • C Program to read employee details (name, salary, address) and print the same using structure. • C-Program to read marks of three students in 3 subjects. Calculate the total marks scored, student wise and subject wise using structure.	
12	Creating and Running C Programs on Files: • C-Program to demonstrate function fread()/fscanf() • C-Program to demonstrate function fwrite()/fprintf()	

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test-I	30	C01, C02, C03
Internal Test-II	30	C03, C04, C05
Average of the two internal test shall be taken for 30 marks		
Other components		
Lab Component Evaluation	20	C01, C02, C03, C04, C05
Semester-End Examination(SEE)	100	C01, C02, C03, C04, C05

Programs

1.Creating and Running Simple C Programs:

- C-Program to calculate the sum of three numbers / C-Program to demonstrate a Simple Calculator.
- C-Program to calculate the area and circumference of a circle using PI as a defined constant.
- C-Program to convert temperature given in Celsius to Fahrenheit and Fahrenheit to Celsius
- C-Program to compute the roots of a quadratic equation by accepting the coefficients.

Sum of three numbers

```
#include <stdio.h>
int main(){
    int a, b, c, sum;
    float avg;

    // Asking for input
    printf("Enter 3 numbers: \n");
    scanf("%d %d %d", &a, &b, &c);

    // Calculating sum
    sum = a + b + c;

    // Displaying output
    printf("Sum = %d \n", sum);
    return 0;
}
```

Output:

```
Enter 3 numbers:
5 6 7
Sum = 18
Average = 6.00
```

C program to demonstrate a simple Calculator

```
#include<stdio.h>

int main() {
    char operator;
    double num1, num2, result;

    printf("Enter an operator (+, -, *, /): ");
    scanf("%c", &operator);

    printf("Enter two operands: ");
    scanf("%lf %lf", &num1, &num2);

    switch (operator) {
        case '+':
            result = num1 + num2;
            printf("Result: %.2lf\n", result);
            break;
        case '-':
            result = num1 - num2;
            printf("Result: %.2lf\n", result);
            break;
        case '*':
            result = num1 * num2;
            printf("Result: %.2lf\n", result);
            break;
        case '/':
            if (num2 != 0) {
                result = num1 / num2;
                printf("Result: %.2lf\n", result);
            } else {
                printf("Error! Division by zero.\n");
            }
            break;
        default:
            printf("Error! Invalid operator.\n");
    }

    return 0;
}
```

Output:

Enter an operator (+, -, *, /): +

Enter two operands: 5 6

Result: 11.00

C-Program to calculate the area and circumference of a circle using PI as a defined constant

```
#include <stdio.h>
```

```
#define PI 3.14159
```

```
int main() {  
    double radius, area, circumference;  
  
    printf("Enter the radius of the circle: ");  
    scanf("%lf", &radius);  
  
    // Calculate the area of the circle  
    area = PI * radius * radius;  
  
    // Calculate the circumference of the circle  
    circumference = 2 * PI * radius;  
  
    printf("Area of the circle: %.2lf\n", area);  
    printf("Circumference of the circle: %.2lf\n", circumference);  
  
    return 0;  
}
```

Output:

Enter the radius of the circle: 6

Area of the circle: 113.10

Circumference of the circle: 37.70

C-Program to convert temperature given in Celsius to Fahrenheit and Fahrenheit to Celsius

```
#include <stdio.h>

double celsiusToFahrenheit(double celsius) {
    return (celsius * 9.0/5.0) + 32.0;
}

double fahrenheitToCelsius(double fahrenheit) {
    return (fahrenheit - 32.0) * 5.0/9.0;
}

int main() {
    char choice;
    double temperature;

    printf("Enter temperature: ");
    scanf("%lf", &temperature);

    printf("Choose conversion:\n");
    printf("a. Celsius to Fahrenheit\n");
    printf("b. Fahrenheit to Celsius\n");
    printf("Enter your choice (a/b): ");
    scanf(" %c", &choice);

    switch (choice) {
        case 'a':
        case 'A':
            printf("%.2lf Celsius is equal to %.2lf Fahrenheit.\n", temperature,
celsiusToFahrenheit(temperature));
            break;
        case 'b':
        case 'B':
            printf("%.2lf Fahrenheit is equal to %.2lf Celsius.\n", temperature,
fahrenheitToCelsius(temperature));
            break;
        default:
            printf("Invalid choice.\n");
    }

    return 0;
}
```

Output:

Enter temperature: 34

Choose conversion:

a. Celsius to Fahrenheit

b. Fahrenheit to Celsius

Enter your choice (a/b): a

34.00 Celsius is equal to 93.20 Fahrenheit.

2. Creating and Running C Programs on Expressions :

- C-Program to calculate quotient and remainder of two numbers.)
- C-Program to evaluate two complex expressions.
- C-Program to demonstrate automatic and type casting of numeric types
- C-Program to calculate the total sales given the unit price, quantity, discount and tax rate
- C-Program to calculate a student's average score for a course with 4 quizzes, 2 midterms and a final. The quizzes are weighted 30%, the midterms 40% and the final 30%.

C-Program to calculate quotient and remainder of two numbers.)

```
#include <stdio.h>
```

```
int main() {  
    int dividend, divisor, quotient, remainder;  
  
    printf("Enter dividend: ");  
    scanf("%d", &dividend);  
  
    printf("Enter divisor: ");  
    scanf("%d", &divisor);  
  
    // Calculate quotient  
    quotient = dividend / divisor;  
  
    // Calculate remainder  
    remainder = dividend % divisor;  
  
    printf("Quotient: %d\n", quotient);  
    printf("Remainder: %d\n", remainder);  
  
    return 0;  
}
```


Output:
Enter dividend: 7
Enter divisor: 3
Quotient: 2
Remainder: 1

C-Program to evaluate two complex expressions.

```
#include <stdio.h>
```

```
typedef struct {  
    double real;  
    double imag;  
} Complex;
```

```
Complex add(Complex num1, Complex num2) {  
    Complex result;  
    result.real = num1.real + num2.real;  
    result.imag = num1.imag + num2.imag;  
    return result;  
}
```

```
Complex multiply(Complex num1, Complex num2) {  
    Complex result;  
    result.real = (num1.real * num2.real) - (num1.imag * num2.imag);  
    result.imag = (num1.real * num2.imag) + (num1.imag * num2.real);  
    return result;  
}
```

```
int main() {  
    Complex num1, num2, sum, product;  
  
    // Input for the first complex number  
    printf("Enter real and imaginary parts of first complex number: ");  
    scanf("%lf %lf", &num1.real, &num1.imag);  
  
    // Input for the second complex number  
    printf("Enter real and imaginary parts of second complex number: ");  
    scanf("%lf %lf", &num2.real, &num2.imag);  
  
    // Calculate sum of complex numbers  
    sum = add(num1, num2);  
  
    // Calculate product of complex numbers  
    product = multiply(num1, num2);
```

```
// Display the sum and product
printf("Sum: %.2lf + %.2lf i\n", sum.real, sum.imag);
printf("Product: %.2lf + %.2lf i\n", product.real, product.imag);

return 0;
}
```

Output:

Enter real and imaginary parts of first complex number: 2

+ 3j

Enter real and imaginary parts of second complex number: Sum: 5.00 + 0.00 i

Product: 6.00 + 0.00 i

C-Program to demonstrate automatic and type casting of numeric types

```
#include <stdio.h>
```

```
int main() {
    // Automatic type conversion (promotion)
    int a = 10;
    double b = 3.5;

    double result_auto = a + b; // int is automatically promoted to double

    printf("Automatic Type Conversion:\n");
    printf("a + b = %.2lf\n", result_auto);

    // Type casting (explicit conversion)
    double x = 20.7;
    int y = (int)x; // double is explicitly casted to int

    printf("\nType Casting:\n");
    printf("x (double) = %.2lf\n", x);
    printf("y (int) = %d\n", y);

    return 0;
}
```

Output:

Automatic Type Conversion:

a + b = 13.50

Type Casting:

x (double) = 20.70

y (int) = 20

C-Program to calculate the total sales given the unit price, quantity, discount and tax rate

```
#include <stdio.h>

int main() {
    double unitPrice, quantity, discount, taxRate;
    double totalSales;

    // Input the unit price, quantity, and tax rate
    printf("Enter unit price: ");
    scanf("%lf", &unitPrice);

    printf("Enter quantity: ");
    scanf("%lf", &quantity);

    printf("Enter discount (as a percentage): ");
    scanf("%lf", &discount);

    printf("Enter tax rate (as a percentage): ");
    scanf("%lf", &taxRate);

    // Calculate total sales
    double discountAmount = (discount / 100) * (unitPrice * quantity);
    double taxableAmount = (unitPrice * quantity) - discountAmount;
    double taxAmount = (taxRate / 100) * taxableAmount;

    totalSales = taxableAmount + taxAmount;

    // Display the total sales
    printf("Total Sales: $%.2lf\n", totalSales);

    return 0;
}
```

Output:

```
Enter unit price: 25
Enter quantity: 2
Enter discount (as a percentage): 10
Enter tax rate (as a percentage): 2
Total Sales: $45.90
```

C-Program to calculate a student's average score for a course with 4 quizzes, 2 midterms and a final. The quizzes are weighted 30%, the midterms 40% and the final 30%.

```
#include <stdio.h>
```

```
int main() {
    double quiz1, quiz2, quiz3, quiz4, midterm1, midterm2, finalExam;
    double quizWeight = 0.3, midtermWeight = 0.4, finalWeight = 0.3;
    double totalScore;

    // Input scores for quizzes, midterms, and final exam
    printf("Enter scores for the 4 quizzes:\n");
    printf("Quiz 1: ");
    scanf("%lf", &quiz1);

    printf("Quiz 2: ");
    scanf("%lf", &quiz2);

    printf("Quiz 3: ");
    scanf("%lf", &quiz3);

    printf("Quiz 4: ");
    scanf("%lf", &quiz4);

    printf("\nEnter scores for the 2 midterms:\n");
    printf("Midterm 1: ");
    scanf("%lf", &midterm1);

    printf("Midterm 2: ");
    scanf("%lf", &midterm2);

    printf("\nEnter score for the final exam: ");
    scanf("%lf", &finalExam);

    // Calculate the weighted scores for each component
    double weightedQuizScore = (quiz1 + quiz2 + quiz3 + quiz4) / 4 * quizWeight;
    double weightedMidtermScore = ((midterm1 + midterm2) / 2) * midtermWeight;
    double weightedFinalScore = finalExam * finalWeight;

    // Calculate the total score
    totalScore = weightedQuizScore + weightedMidtermScore + weightedFinalScore;

    // Display the total score
    printf("\nTotal Score: %.2lf\n", totalScore);

    return 0;
}
```

```
}
```

Output:

Enter scores for the 4 quizzes:

Quiz 1: 56

Quiz 2: 78

Quiz 3: 67

Quiz 4: 89

Enter scores for the 2 midterms:

Midterm 1: 56

Midterm 2: 78

Enter score for the final exam: 89

Total Score: 75.25

3.Creating and Running C Programs on Making Decision:

- C-Program to determine the use of the character classification functions found in c-type library.
- C-Program to read a test score, calculate the grade for the score and print the grade.
- C-Program to uses a menu to allow the user to add, multiply, subtract and divide two numbers using switch case.
- C-Program to read the name of the user, number of units consumed and print out the charges.

An electricity board charges the following rates for the use of electricity:

- For the first 200 units 80 paise per unit
- For the next 100 units 90 paise per unit
- Beyond 300 units Rs 1 per unit.

All users are charged a minimum of Rs. 100 as meter charge. If the total amount is more than Rs 400, then an additional surcharge of 15% of total amount is charged

C-Program to determine the use of the character classification functions found in c-type library.

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
int main() {
```

```
    char ch;
```

```
    printf("Enter a character: ");
```

```
    scanf(" %c", &ch); // Note the space before %c to skip whitespace characters
```

```
    // Check if the character is a digit
```

```
    if (isdigit(ch))
```

```
        printf("%c is a digit.\n", ch);
```

```
else
    printf("%c is not a digit.\n", ch);

// Check if the character is an alphabetic character
if (isalpha(ch))
    printf("%c is an alphabetic character.\n", ch);
else
    printf("%c is not an alphabetic character.\n", ch);

// Check if the character is a whitespace character
if (isspace(ch))
    printf("%c is a whitespace character.\n", ch);
else
    printf("%c is not a whitespace character.\n", ch);

// Check if the character is an uppercase letter
if (isupper(ch))
    printf("%c is an uppercase letter.\n", ch);
else
    printf("%c is not an uppercase letter.\n", ch);

// Check if the character is a lowercase letter
if (islower(ch))
    printf("%c is a lowercase letter.\n", ch);
else
    printf("%c is not a lowercase letter.\n", ch);

return 0;
}
```

Output:

Enter a character: t

t is not a digit.

t is an alphabetic character.

t is not a whitespace character.

t is not an uppercase letter.

t is a lowercase letter.

C-Program to read a test score, calculate the grade for the score and print the grade.

```
#include <stdio.h>

int main() {
    int score;

    printf("Enter the test score (out of 100): ");
    scanf("%d", &score);

    // Determine the grade based on the score
    char grade;
    if (score >= 90) {
        grade = 'A';
    } else if (score >= 80) {
        grade = 'B';
    } else if (score >= 70) {
        grade = 'C';
    } else if (score >= 60) {
        grade = 'D';
    } else {
        grade = 'F';
    }

    printf("Grade for the test score %d is: %c\n", score, grade);

    return 0;
}
```

Output:

Enter the test score (out of 100): 98
Grade for the test score 98 is: A

C-Program to uses a menu to allow the user to add, multiply, subtracts and divides two numbers using switch case

```
#include <stdio.h>
```

```
int main() {
    int choice;
    double num1, num2, result;

    printf("Enter first number: ");
    scanf("%lf", &num1);

    printf("Enter second number: ");
    scanf("%lf", &num2);

    printf("Menu:\n");
    printf("1. Add\n");
    printf("2. Subtract\n");
    printf("3. Multiply\n");
    printf("4. Divide\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            result = num1 + num2;
            printf("Result of addition: %.2lf\n", result);
            break;
        case 2:
            result = num1 - num2;
            printf("Result of subtraction: %.2lf\n", result);
            break;
        case 3:
            result = num1 * num2;
            printf("Result of multiplication: %.2lf\n", result);
            break;
        case 4:
            if (num2 != 0) {
                result = num1 / num2;
                printf("Result of division: %.2lf\n", result);
            } else {
                printf("Error! Division by zero.\n");
            }
            break;
        default:
            printf("Invalid choice!\n");
    }
}
```



```
    return 0;
}
```

Output:

Enter first number: 4

Enter second number: 6

Menu:

1. Add

2. Subtract

3. Multiply

4. Divide

Enter your choice: 1

Result of addition: 10.00

C-Program to read the name of the user, number of units consumed and print out the charges. An electricity board charges the following rates for the use of electricity:

- **For the first 200 units 80 paise per unit**
- **For the next 100 units 90 paise per unit**
- **Beyond 300 units Rs 1 per unit.**

```
#include <stdio.h>
```

```
int main() {
    char name[100];
    int units;
    double totalCharge;
```

```
    // Input user's name and units consumed
```

```
    printf("Enter your name: ");
```

```
    scanf("%s", name);
```

```
    printf("Enter the number of units consumed: ");
```

```
    scanf("%d", &units);
```

```
    // Calculate charges based on rates
```

```
    if (units <= 200) {
```

```
        totalCharge = units * 0.80; // 80 paise per unit
```

```
    } else if (units <= 300) {
```

```
        totalCharge = 200 * 0.80 + (units - 200) * 0.90; // 80 paise for first 200 units, 90 paise for
next 100 units
```

```
    } else {
```

```
        totalCharge = 200 * 0.80 + 100 * 0.90 + (units - 300) * 1.0; // Rs 1 per unit beyond 300
units
```

```
}

// Display the charges
printf("Hello, %s!\n", name);
printf("Total charges for %d units: Rs %.2lf\n", units, totalCharge);

return 0;
}
```

Output:

```
Enter your name: nithin
Enter the number of units consumed: 45
Hello, nithin!
Total charges for 45 units: Rs 36.00
```

4.Creating and Running C Programs on Repetition or Loops:

- **C-Program to print a number series from 1 to a user-specified limit in the form of a angle**
- **C-Program to print the number and sum of digits in an integer.**
- **C-Program to calculate the factorial of a number using for loop/ Recursion**
- **C-Program to calculate nth Fibonacci number.**
- **C-Program to convert binary to a decimal number**

C-Program to print a number series from 1 to a user-specified limit in the form of a angle

```
#include <stdio.h>

int main() {
    int limit;

    printf("Enter the limit: ");
    scanf("%d", &limit);

    for (int i = 1; i <= limit; i++) {
        printf("%d degrees\n", i);
    }

    return 0;
}
```

Output:

```
Enter the limit: 6
1 degrees
2 degrees
3 degrees
4 degrees
5 degrees
6 degrees
```

C-Program to print the number and sum of digits in an integer

```
#include <stdio.h>

int main() {
    int num, originalNum, sum = 0, remainder;

    printf("Enter an integer: ");
    scanf("%d", &num);

    originalNum = num;

    while (num != 0) {
        remainder = num % 10;
        sum += remainder;
        num /= 10;
    }

    printf("Sum of digits of %d = %d\n", originalNum, sum);

    return 0;
}
```

Output:

Enter an integer: 345

Sum of digits of 345 = 12

C-Program to calculate the factorial of a number using for loop/ Recursion

```
#include <stdio.h>

int main() {
    int num;
    unsigned long long factorial = 1;

    printf("Enter a non-negative integer: ");
    scanf("%d", &num);

    for (int i = 1; i <= num; ++i) {
        factorial *= i;
    }
    printf("Factorial of %d = %llu\n", num, factorial);

    return 0;
}
```

Output:

Enter a non-negative integer: 5

Factorial of 5 = 120

C-Program to calculate nth Fibonacci number.

```
#include <stdio.h>
```

```
int main() {  
    int n, i, a = 0, b = 1, next;  
  
    printf("Enter the number of terms: ");  
    scanf("%d", &n);  
  
    printf("Fibonacci Series: ");  
  
    for (i = 1; i <= n; ++i) {  
        printf("%d, ", a);  
        next = a + b;  
        a = b;  
        b = next;  
    }  
  
    return 0;  
}
```

Output:

Enter the number of terms: 4

Fibonacci Series: 0, 1, 1, 2,

C-Program to convert binary to a decimal number

```
#include <stdio.h>
```

```
int binaryToDecimal(int binary) {  
    int decimal = 0, base = 1, remainder;  
  
    while (binary > 0) {  
        remainder = binary % 10;  
        decimal += remainder * base;  
        binary /= 10;  
        base *= 2;  
    }  
  
    return decimal;  
}  
  
int main() {  
    int binary;  
  
    printf("Enter a binary number: ");  
    scanf("%d", &binary);  
  
    printf("Decimal equivalent: %d\n", binaryToDecimal(binary));  
  
    return 0;  
}
```

Output:

Enter a binary number: 1001

Decimal equivalent: 9

5. Creating and Running C Programs on One Dimensional Arrays:

- **C-Program to print square of index and print it.**
- **C-Program to calculate average of the number in an array.**
- **C-Program to sort the list using bubble sort.**
- **C-Program to search an ordered list using binary search.**

-Program to print square of index and print it.

```
#include <stdio.h>
```

```
int main() {  
    int n;  
  
    printf("Enter the number of elements: ");  
    scanf("%d", &n);  
  
    for (int i = 0; i < n; i++) {  
        printf("Square of index %d: %d\n", i, i * i);  
    }  
  
    return 0;  
}
```

Output:

```
Enter the number of elements: 4  
Square of index 0: 0  
Square of index 1: 1  
Square of index 2: 4  
Square of index 3: 9
```

C-Program to calculate average of the number in an array.

```
#include <stdio.h>
```

```
int main() {  
    int n;  
    double sum = 0.0;  
  
    printf("Enter the number of elements: ");  
    scanf("%d", &n);  
  
    int arr[n];  
  
    printf("Enter the elements:\n");  
    for (int i = 0; i < n; i++) {  
        scanf("%d", &arr[i]);  
        sum += arr[i];  
    }
```

```

    }

    double average = sum / n;
    printf("Average: %.2lf\n", average);

    return 0;
}

```

Output:

Enter the number of elements: 5

Enter the elements:

3 4 5 6 7

Average: 5.00

C-Program to sort the list using bubble sort.

```
#include <stdio.h>
```

```

void bubbleSort(int arr[], int n)
{
    for (int i = 0; i < n - 1; i++)
    {
        for (int j = 0; j < n - i - 1; j++)
        {
            if (arr[j] > arr[j + 1]) {
                // Swap arr[j] and arr[j + 1]
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

int main() {
    int n;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter the elements:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
}

```

```

// Sort the array
bubbleSort(arr, n);

printf("Sorted array:\n");
for (int i = 0; i < n; i++) {
    printf("%d ", arr[i]);
}

return 0;
}

```

Output:

Enter the number of elements: 5

Enter the elements:

3 1 2 5 6

Sorted array:

1 2 3 5 6

C-Program to search an ordered list using binary search.

```

#include <stdio.h>
int binarySearch(int arr[], int n, int key) {
    int left = 0;
    int right = n - 1;

    while (left <= right) {
        int mid = left + (right - left) / 2;

        if (arr[mid] == key)
            return mid;

        if (arr[mid] < key)
            left = mid + 1;
        else
            right = mid - 1;
    }

    return -1;
}

int main() {
    int n, key;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

```



```
int arr[n];

printf("Enter the elements in sorted order:\n");
for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
}

printf("Enter the element to search: ");
scanf("%d", &key);

int result = binarySearch(arr, n, key);

if (result == -1)
    printf("Element not found.\n");
else
    printf("Element found at index %d.\n", result);

return 0;
}
```

Output:

```
Enter the number of elements: 4
Enter the elements in sorted order:
4
5
6
7
Enter the element to search: 7
Element found at index 3.
```

6.Creating and Running C Programs on Two Dimensional Arrays:

- **C-Program to perform addition of two matrices.**
- **C-Program to perform multiplication of two matrices.**
- **C-Program to find transpose of the given matrices.**
- **C-Program to find row sum and column sum and sum of all elements in a matrix.**
- C-Program initialize/fill all the diagonal elements of a matrix with zero and print

C-Program to perform addition of two matrices.

```
#include <stdio.h>
```

```
void addMatrices(int mat1[10][10], int mat2[10][10], int result[10][10], int rows, int cols)
{
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            result[i][j] = mat1[i][j] + mat2[i][j];
        }
    }
}
```

```
void displayMatrix(int mat[10][10], int rows, int cols) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            printf("%d ", mat[i][j]);
        }
        printf("\n");
    }
}
```

```
int main() {
    int mat1[10][10], mat2[10][10], result[10][10];
    int rows, cols;

    printf("Enter the number of rows: ");
    scanf("%d", &rows);

    printf("Enter the number of columns: ");
    scanf("%d", &cols);

    printf("Enter elements of the first matrix:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            scanf("%d", &mat1[i][j]);
        }
    }
}
```

```
printf("Enter elements of the second matrix:\n");
for (int i = 0; i < rows; i++) {
    for (int j = 0; j < cols; j++) {
        scanf("%d", &mat2[i][j]);
    }
}

// Perform matrix addition
addMatrices(mat1, mat2, result, rows, cols);

// Display the result
printf("Sum of the matrices:\n");
displayMatrix(result, rows, cols);

return 0;
}
```

Output:

Enter the number of rows: 2

Enter the number of columns: 2

Enter elements of the first matrix:

2 2 2 2

Enter elements of the second matrix:

2 2 2 2

Sum of the matrices:

4 4

4 4

C-Program to perform multiplication of two matrix

```
#include <stdio.h>
```

```
void multiplyMatrices(int mat1[10][10], int mat2[10][10], int result[10][10], int rows1, int cols1,
int rows2, int cols2) {
    for (int i = 0; i < rows1; i++) {
        for (int j = 0; j < cols2; j++) {
            result[i][j] = 0;
            for (int k = 0; k < cols1; k++) {
                result[i][j] += mat1[i][k] * mat2[k][j];
            }
        }
    }
}
```

```
void displayMatrix(int mat[10][10], int rows, int cols) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            printf("%d ", mat[i][j]);
        }
        printf("\n");
    }
}
```

```
int main() {
    int mat1[10][10], mat2[10][10], result[10][10];
    int rows1, cols1, rows2, cols2;

    printf("Enter the number of rows for matrix 1: ");
    scanf("%d", &rows1);

    printf("Enter the number of columns for matrix 1: ");
    scanf("%d", &cols1);

    printf("Enter elements of matrix 1:\n");
    for (int i = 0; i < rows1; i++) {
        for (int j = 0; j < cols1; j++) {
            scanf("%d", &mat1[i][j]);
        }
    }

    printf("Enter the number of rows for matrix 2: ");
    scanf("%d", &rows2);

    printf("Enter the number of columns for matrix 2: ");
    scanf("%d", &cols2);
```

```

printf("Enter elements of matrix 2:\n");
for (int i = 0; i < rows2; i++) {
    for (int j = 0; j < cols2; j++) {
        scanf("%d", &mat2[i][j]);
    }
}

// Check if matrices can be multiplied
if (cols1 != rows2) {
    printf("Matrices cannot be multiplied.\n");
    return 0;
}

// Perform matrix multiplication
multiplyMatrices(mat1, mat2, result, rows1, cols1, rows2, cols2);

// Display the result
printf("Result of matrix multiplication:\n");
displayMatrix(result, rows1, cols2);

return 0;
}

```

Output:

```

Enter the number of rows for matrix 1: 2
Enter the number of columns for matrix 1: 2
Enter elements of matrix 1:
2 2 2 2
Enter the number of rows for matrix 2: 2
Enter the number of columns for matrix 2: 2
Enter elements of matrix 2:
2 2 2 2
Result of matrix multiplication:
8 8
8 8

```

C-Program to find transpose of the given matrices.

```
#include <stdio.h>

void transposeMatrix(int mat[10][10], int transpose[10][10], int rows, int cols) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            transpose[j][i] = mat[i][j];
        }
    }
}

void displayMatrix(int mat[10][10], int rows, int cols) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            printf("%d ", mat[i][j]);
        }
        printf("\n");
    }
}

int main() {
    int mat[10][10], transpose[10][10];
    int rows, cols;

    printf("Enter the number of rows: ");
    scanf("%d", &rows);

    printf("Enter the number of columns: ");
    scanf("%d", &cols);

    printf("Enter elements of the matrix:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            scanf("%d", &mat[i][j]);
        }
    }

    // Calculate the transpose of the matrix
    transposeMatrix(mat, transpose, rows, cols);

    // Display the original matrix
    printf("Original Matrix:\n");
    displayMatrix(mat, rows, cols);
```

```

// Display the transpose
printf("\nTranspose of the Matrix:\n");
displayMatrix(transpose, cols, rows);

return 0;
}

```

Output:

```

Enter the number of rows: 2
Enter the number of columns: 2
Enter elements of the matrix:
2 3 4 5
Original Matrix:
2 3
4 5

```

```

Transpose of the Matrix:
2 4
3 5

```

C-Program to find row sum and column sum and sum of all elements in a matrix

```

#include <stdio.h>
#include <stdio.h>

void calculateSums(int mat[10][10], int rows, int cols) {
    int rowSum[10] = {0};
    int colSum[10] = {0};
    int totalSum = 0;

    // Calculate row sums and total sum
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            rowSum[i] += mat[i][j];
            totalSum += mat[i][j];
        }
    }

    // Calculate column sums
    for (int j = 0; j < cols; j++) {
        for (int i = 0; i < rows; i++) {
            colSum[j] += mat[i][j];
        }
    }
}

```

```

// Display row sums
printf("Row sums:\n");
for (int i = 0; i < rows; i++) {
    printf("Row %d: %d\n", i + 1, rowSum[i]);
}

// Display column sums
printf("\nColumn sums:\n");
for (int j = 0; j < cols; j++) {
    printf("Column %d: %d\n", j + 1, colSum[j]);
}

// Display total sum
printf("\nTotal sum of all elements: %d\n", totalSum);
}

void displayMatrix(int mat[10][10], int rows, int cols) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            printf("%d ", mat[i][j]);
        }
        printf("\n");
    }
}

int main() {
    int mat[10][10];
    int rows, cols;

    printf("Enter the number of rows: ");
    scanf("%d", &rows);

    printf("Enter the number of columns: ");
    scanf("%d", &cols);

    printf("Enter elements of the matrix:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            scanf("%d", &mat[i][j]);
        }
    }

    // Display the original matrix
    printf("\nOriginal Matrix:\n");
    displayMatrix(mat, rows, cols);
}

```



```
// Calculate and display row sum, column sum, and total sum
printf("\n\nCalculating Sums...\n");
calculateSums(mat, rows, cols);

return 0;
}
```

Output:

Enter the number of rows: 2

Enter the number of columns: 2

Enter elements of the matrix:

2 2 2 2

Original Matrix:

2 2

2 2

Calculating Sums...

Row sums:

Row 1: 4

Row 2: 4

Column sums:

Column 1: 4

Column 2: 4

Total sum of all elements: 8

C-Program initialize/fill all the diagonal elements of a matrix with zero and print

```

#include <stdio.h>
void fillDiagonalWithZero(int mat[10][10], int rows, int cols) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            if (i == j) {
                mat[i][j] = 0;
            }
        }
    }
}

void displayMatrix(int mat[10][10], int rows, int cols) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            printf("%d ", mat[i][j]);
        }
        printf("\n");
    }
}

int main() {
    int mat[10][10];
    int rows, cols;

    printf("Enter the number of rows: ");
    scanf("%d", &rows);

    printf("Enter the number of columns: ");
    scanf("%d", &cols);

    printf("Enter elements of the matrix:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            scanf("%d", &mat[i][j]);
        }
    }
    // Fill diagonal with zero
    fillDiagonalWithZero(mat, rows, cols);

    // Display the modified matrix
    printf("\nMatrix with diagonal elements as zero:\n");
    displayMatrix(mat, rows, cols);

    return 0;
}

```

```
}
```

Output:

Enter the number of rows: 2

Enter the number of columns: 2

Enter elements of the matrix:

2 2 2 2

Matrix with diagonal elements as zero:

0 2

2 0

7.Creating and Running C Programs on User Defined Functions:

- C-program to read a number, Find its factorial using function with argument and with return type.
- C-Program to read two number, Find its GCD and LCM using function with arguments and without return type.
- C-Program to read a number, Find whether it is a palindrome or not using function without argument and with return type.
- C-Program to read a number, Find whether it is prime number or not using function without arguments and without return type.

C-program to read a number, Find its factorial using function with argument and with return type.

```
#include <stdio.h>
unsigned long long factorial(int num) {
    if (num == 0 || num == 1)
        return 1;
    else
        return num * factorial(num - 1);
}

int main() {
    int num;

    printf("Enter a non-negative integer: ");
    scanf("%d", &num);

    if (num < 0) {
        printf("Factorial is not defined for negative numbers.\n");
    } else {
        unsigned long long result = factorial(num);
        printf("Factorial of %d = %llu\n", num, result);
    }
    return 0;
}
```

Output:

Enter a non-negative integer: 6

Factorial of 6 = 720

C-Program to read two number, Find its GCD and LCM using function with arguments and without return type.

```
#include <stdio.h>
```

```
void calculateGCDandLCM(int num1, int num2, int *gcd, int *lcm) {
```

```
    int i, max;
```

```
    // Find maximum between the two numbers
```

```
    max = (num1 > num2) ? num1 : num2;
```

```
    // Find GCD
```

```
    for (i = 1; i <= max; i++) {
```

```
        if (num1 % i == 0 && num2 % i == 0)
```

```
            *gcd = i;
```

```
    }
```

```
    // Find LCM
```

```
    *lcm = (num1 * num2) / *gcd;
```

```
}
```

```
int main() {
```

```
    int num1, num2, gcd, lcm;
```

```
    printf("Enter first number: ");
```

```
    scanf("%d", &num1);
```

```
    printf("Enter second number: ");
```

```
    scanf("%d", &num2);
```

```
    // Calculate GCD and LCM
```

```
    calculateGCDandLCM(num1, num2, &gcd, &lcm);
```

```
    printf("GCD of %d and %d is: %d\n", num1, num2, gcd);
```

```
    printf("LCM of %d and %d is: %d\n", num1, num2, lcm);
```

```
    return 0;
```

```
}
```

Output:

Enter first number: 4

Enter second number: 6

GCD of 4 and 6 is: 2

LCM of 4 and 6 is: 12

C-Program to read a number, Find whether it is a palindrome or not using function without argument and with return type

```
#include <stdio.h>
```

```
int isPalindrome() {
    int num, reversedNum = 0, originalNum;

    printf("Enter a number: ");
    scanf("%d", &num);

    originalNum = num;

    // Reverse the number
    while (num != 0) {
        int remainder = num % 10;
        reversedNum = reversedNum * 10 + remainder;
        num /= 10;
    }

    // Check if the original number and the reversed number are the same
    if (originalNum == reversedNum)
        return 1; // The number is a palindrome
    else
        return 0; // The number is not a palindrome
}

int main() {
    int result;

    result = isPalindrome();

    if (result)
        printf("The number is a palindrome.\n");
    else
        printf("The number is not a palindrome.\n");

    return 0;
}
```

Output:

Enter a number: 121

The number is a palindrome.

Enter a number: 132

The number is not a palindrome.

C-Program to read a number, Find whether it is prime number or not using function without arguments and without return type.

```
#include <stdio.h>
```

```
void checkPrime() {  
    int num, i, flag = 0;
```

```
    printf("Enter a number: ");  
    scanf("%d", &num);
```

```
    // 0 and 1 are not prime numbers  
    if (num == 0 || num == 1) {  
        printf("%d is not a prime number.\n", num);  
        return;  
    }
```

```
    for (i = 2; i <= num / 2; ++i) {  
        if (num % i == 0) {  
            flag = 1;  
            break;  
        }  
    }
```

```
    if (flag == 0)  
        printf("%d is a prime number.\n", num);  
    else  
        printf("%d is not a prime number.\n", num);  
}
```

```
int main() {  
    checkPrime();  
    return 0;  
}
```

Output:

Enter a number: 4

4 is not a prime number.

Enter a number: 5

5 is a prime number.

8.Creating and Running C Programs on Strings:

- **C-program read two strings, Combine them without using string built-in functions.**
- **C-program read two strings, Compare them without using string built-in functions.**
- **C-program read two strings, concatenate them without using string built-in functions.**
- **C Program to Check if the Substring is Present in the Given String.**
- **C-program to demonstrate built-in sting functions like strlen(), strcpy(), strcmp(), strcat()**

C-program read two strings, Combine them without using string built-in functions.

```
#include <stdio.h>
```

```
void combineStrings(char str1[], char str2[], char combined[]) {  
    int i, j;
```

```
    // Copy the first string to the combined string  
    for (i = 0; str1[i] != '\0'; ++i) {  
        combined[i] = str1[i];  
    }
```

```
    // Concatenate the second string to the combined string  
    for (j = 0; str2[j] != '\0'; ++j) {  
        combined[i + j] = str2[j];  
    }
```

```
    // Null-terminate the combined string  
    combined[i + j] = '\0';  
}
```

```
int main() {  
    char str1[100], str2[100], combined[200];
```

```
    printf("Enter the first string: ");  
    scanf("%s", str1);
```

```
    printf("Enter the second string: ");
```

```

scanf("%s", str2);

combineStrings(str1, str2, combined);

printf("Combined string: %s\n", combined);

return 0;
}

```

Output:

```

Enter the first string: hello
Enter the second string: good
Combined string: hellogood

```

C-program read two strings, Compare them without using string built-in functions

```

#include <stdio.h>

int compareStrings(char str1[], char str2[]) {
    int i = 0;

    while (str1[i] != '\0' && str2[i] != '\0') {
        if (str1[i] != str2[i]) {
            return str1[i] - str2[i];
        }
        i++;
    }

    return str1[i] - str2[i];
}

int main() {
    char str1[100], str2[100];

    printf("Enter the first string: ");
    scanf("%s", str1);

    printf("Enter the second string: ");
    scanf("%s", str2);

    int result = compareStrings(str1, str2);

    if (result == 0) {
        printf("Strings are equal.\n");
    } else if (result < 0) {

```



```

    printf("First string is less than the second string.\n");
} else {
    printf("First string is greater than the second string.\n");
}

return 0;
}

```

Output:

Enter the first string: hello

Enter the second string: good

First string is greater than the second string.

C Program to Check if the Substring is Present in the Given String

```
#include<stdio.h>
```

```

int main()
{
    char str[80], search[10];
    int count1 = 0, count2 = 0, i, j, flag;

    printf("Enter a string:");
    gets(str);
    printf("Enter search substring:");
    gets(search);
    while (str[count1] != '\0')
        count1++;
    while (search[count2] != '\0')
        count2++;
    for (i = 0; i <= count1 - count2; i++)
    {
        for (j = i; j < i + count2; j++)
        {
            flag = 1;
            if (str[j] != search[j - i])
            {
                flag = 0;
                break;
            }
        }
        if (flag == 1)
            break;
    }
    if (flag == 1)
        printf("SEARCH SUCCESSFUL!");
    else

```

```

    printf("SEARCH UNSUCCESSFUL!");

    return 0;
}

```

Output

```

Enter a string:HELLO HI
Enter search substring:HI
SEARCH SUCCESSFUL!

```

C-program to demonstrate built-in string functions like strlen(), strcpy(), strcmp(), strcat()

```

#include <stdio.h>
#include <string.h>

int main() {
    char str1[100], str2[100];

    // Demonstrate strlen()
    printf("Enter a string: ");
    scanf("%s", str1);
    printf("Length of the string: %lu\n", strlen(str1));

    // Demonstrate strcpy()
    strcpy(str2, str1);
    printf("Copied string using strcpy(): %s\n", str2);

    // Demonstrate strcmp()
    printf("Enter another string: ");
    scanf("%s", str2);
    int cmpResult = strcmp(str1, str2);
    if (cmpResult == 0)
        printf("Strings are equal.\n");
    else if (cmpResult < 0)
        printf("First string is less than the second string.\n");
    else
        printf("First string is greater than the second string.\n");

    // Demonstrate strcat()
    strcat(str1, str2);
    printf("Concatenated string using strcat(): %s\n", str1);

    return 0;
}

```

Output:

/tmp/iEaAHSxUVB.o

Enter a string: hello

Length of the string: 5

Copied string using strcpy(): hello

Enter another string: good

First string is greater than the second string.

Concatenated string using strcat(): hellogood

9.Creating and Running C Programs on Storage Classes and Pointers:

- **C-program to show the use of auto and static variable.**
- **C-program to add two numbers using pointers. / C-program to swap two numbers using pointers.**
- **C-program to show how the same pointer can point to different data variable.**
- / C-program to show the use of different pointers point to the same variable**
- **C-Program to read an array of elements, Compute its sum using pointers.**

C-program to show the use of auto and static variable.

```
#include <stdio.h>
```

```
void autoExample() {  
    auto int x = 10; // auto variable  
  
    printf("Auto variable: %d\n", x);  
  
    x++; // Modify the auto variable  
  
    printf("Modified auto variable: %d\n", x);  
}
```

```
void staticExample() {  
    static int y = 20; // static variable  
  
    printf("Static variable: %d\n", y);  
  
    y++; // Modify the static variable  
  
    printf("Modified static variable: %d\n", y);  
}
```

```
int main() {  
    printf("Inside main function:\n");  
  
    autoExample();  
}
```

```
staticExample();

printf("Again inside main function:\n");

autoExample();
staticExample();

return 0;
}
```

Output:

Inside main function:

Auto variable: 10

Modified auto variable: 11

Static variable: 20

Modified static variable: 21

Again inside main function:

Auto variable: 10

Modified auto variable: 11

Static variable: 21

Modified static variable: 22

C-program to add two numbers using pointers. / C-program to swap two numbers using pointers

C-program to add two numbers using pointers

```
#include <stdio.h>

void addNumbers(int *a, int *b, int *sum) {
    *sum = *a + *b;
}

int main() {
    int num1, num2, sum;

    printf("Enter first number: ");
    scanf("%d", &num1);

    printf("Enter second number: ");
    scanf("%d", &num2);

    // Passing addresses of num1 and num2 to addNumbers function
    addNumbers(&num1, &num2, &sum);

    printf("Sum: %d + %d = %d\n", num1, num2, sum);
}
```

```
    return 0;  
}
```

Output

Enter first number: 5

Enter second number: 6

Sum: 5 + 6 = 11

C-program to swap two numbers using pointers

```
#include <stdio.h>
```

```
void swapNumbers(int *a, int *b) {  
    int temp = *a;  
    *a = *b;  
    *b = temp;  
}
```

```
int main() {  
    int num1, num2;  
  
    printf("Enter first number: ");  
    scanf("%d", &num1);  
  
    printf("Enter second number: ");  
    scanf("%d", &num2);  
  
    printf("Before swapping: num1 = %d, num2 = %d\n", num1, num2);  
  
    // Passing addresses of num1 and num2 to swapNumbers function  
    swapNumbers(&num1, &num2);  
  
    printf("After swapping: num1 = %d, num2 = %d\n", num1, num2);  
  
    return 0;  
}
```

Output:

Enter first number: 4

Enter second number: 6

Before swapping: num1 = 4, num2 = 6

After swapping: num1 = 6, num2 = 4

C-program to show how the same pointer can point to different data variable

```
#include <stdio.h>
```

```
int main() {  
    int a = 10, b = 20;  
    int *ptr;  
  
    ptr = &a; // ptr points to variable a  
    printf("Value at ptr: %d\n", *ptr); // prints 10  
  
    ptr = &b; // ptr now points to variable b  
    printf("Value at ptr: %d\n", *ptr); // prints 20  
  
    return 0;  
}
```

Output:

Value at ptr: 10

Value at ptr: 20

C-program to show the use of different pointers point to the same variable

```
#include <stdio.h>
```

```
int main() {  
    int x = 10;  
    int *ptr1, *ptr2;  
    ptr1 = &x; // ptr1 points to variable x  
    ptr2 = &x; // ptr2 also points to variable x  
  
    printf("Value at ptr1: %d\n", *ptr1); // prints 10  
    printf("Value at ptr2: %d\n", *ptr2); // prints 10  
  
    return 0;  
}
```

Output:

Value at ptr1: 10

Value at ptr2: 10

C-Program to read an array of elements, Compute its sum using pointers.

```
#include <stdio.h>

int main() {
    int arr[100], n, sum = 0;

    printf("Enter the number of elements (max 100): ");
    scanf("%d", &n);

    printf("Enter elements of the array:\n");
    for (int i = 0; i < n; i++) {
        printf("Enter element %d: ", i + 1);
        scanf("%d", &arr[i]);
    }

    // Compute sum using pointers
    int *ptr = arr;
    for (int i = 0; i < n; i++) {
        sum += *ptr;
        ptr++;
    }

    printf("Sum of the array elements: %d\n", sum);

    return 0;
}
```

Output:

```
Enter the number of elements (max 100): 7
Enter elements of the array:
Enter element 1: 4
Enter element 2: 6
Enter element 3: 7
Enter element 4: 8
Enter element 5: 9
Enter element 6: 2
Enter element 7: 1
Sum of the array elements: 37
```

10.Creating and Running C Programs on Derived Types and Unions:

- **C-Program to print selected TV stations for our cable TV systems.**
- **C-Program to demonstrate union of short int and two char**

C-Program to print selected TV stations for our cable TV systems.

```
#include <stdio.h>

int main() {
    char tvStations[10][30] = {
        "CNN",
        "BBC News",
        "ESPN",
        "Discovery Channel",
        "National Geographic",
        "Cartoon Network",
        "HBO",
        "FOX News",
        "CNBC",
        "MTV"
    };

    printf("TV Stations for Cable TV Systems:\n");
    for (int i = 0; i < 10; i++) {
        printf("%d. %s\n", i + 1, tvStations[i]);
    }

    return 0;
}
```

Output:

TV Stations for Cable TV Systems:

1. CNN
2. BBC News
3. ESPN
4. Discovery Channel
5. National Geographic
6. Cartoon Network
7. HBO
8. FOX News
9. CNBC
10. MTV

C-Program to demonstrate union of short int and two char

```
#include <stdio.h>

union SampleUnion {
    short int s;
    char c[2];
};

int main() {
    union SampleUnion u;

    printf("Enter a short integer: ");
    scanf("%hi", &u.s);

    printf("Short int: %hi\n", u.s);
    printf("Characters (as bytes):\n");
    printf("First char: %c\n", u.c[0]);
    printf("Second char: %c\n", u.c[1]);

    return 0;
}
```

Output:

```
Enter a short integer: 345
Short int: 345
Characters (as bytes):
First char: Y
Second char:
```

11.Creating and Running C Programs on Structures:

- C Program to read employee details (name, salary, address) and print the same using structure.
- C-Program to read marks of three students in 3 subjects. Calculate the total marks scored, student wise and subject wise using structure.

C Program to read employee details (name, salary, address) and print the same using structure

```
#include <stdio.h>

// Define a structure for employee details
struct Employee {
    char name[50];
    float salary;
    char address[100];
};
```

```
};

int main() {
    struct Employee emp;

    printf("Enter employee details:\n");

    printf("Enter name: ");
    scanf(" %[^\\n]", emp.name); // Read the name with spaces

    printf("Enter salary: ");
    scanf("%f", &emp.salary);

    printf("Enter address: ");
    scanf(" %[^\\n]", emp.address); // Read the address with spaces

    // Print employee details
    printf("\nEmployee Details:\n");
    printf("Name: %s\\n", emp.name);
    printf("Salary: %.2f\\n", emp.salary);
    printf("Address: %s\\n", emp.address);

    return 0;
}
```

Output:

```
Enter employee details:
Enter name: nithin
Enter salary: 456789
Enter address: mathikere
Employee Details:
Name: nithin
Salary: 456789.00
Address: mathikere
```

C-Program to read marks of three students in 3 subjects. Calculate the total marks scored, student wise and subject wise using structure.

```
#include <stdio.h>
#define NUM_STUDENTS 3
#define NUM_SUBJECTS 3

// Define a structure for student details
struct Student {
    int marks[NUM_SUBJECTS];
    int totalMarks;
};

int main() {
    struct Student students[NUM_STUDENTS] = {{0}, {0}, {0}};

    // Read marks for each student and subject
    for (int i = 0; i < NUM_STUDENTS; i++) {
        printf("Enter marks for Student %d:\n", i + 1);
        for (int j = 0; j < NUM_SUBJECTS; j++) {
            printf("Enter marks for Subject %d: ", j + 1);
            scanf("%d", &students[i].marks[j]);
            students[i].totalMarks += students[i].marks[j];
        }
    }

    // Print total marks scored by each student
    printf("\nTotal marks scored by each student:\n");
    for (int i = 0; i < NUM_STUDENTS; i++) {
        printf("Student %d Total Marks: %d\n", i + 1, students[i].totalMarks);
    }

    // Calculate subject-wise total marks
    int subjectTotal[NUM_SUBJECTS] = {0};
    for (int i = 0; i < NUM_STUDENTS; i++) {
        for (int j = 0; j < NUM_SUBJECTS; j++) {
            subjectTotal[j] += students[i].marks[j];
        }
    }

    // Print total marks scored in each subject
    printf("\nTotal marks scored in each subject:\n");
    for (int j = 0; j < NUM_SUBJECTS; j++) {
        printf("Subject %d Total Marks: %d\n", j + 1, subjectTotal[j]);
    }

    return 0;
}
```

Output:

Enter marks for Student 1:
Enter marks for Subject 1: 56
Enter marks for Subject 2: 78
Enter marks for Subject 3: 89
Enter marks for Student 2:
Enter marks for Subject 1: 87
Enter marks for Subject 2: 98
Enter marks for Subject 3: 79
Enter marks for Student 3:
Enter marks for Subject 1: 89
Enter marks for Subject 2: 67
Enter marks for Subject 3: 87
Total marks scored by each student:
Student 1 Total Marks: 223
Student 2 Total Marks: 264
Student 3 Total Marks: 243

Total marks scored in each subject:
Subject 1 Total Marks: 232
Subject 2 Total Marks: 243
Subject 3 Total Marks: 255

12.Creating and Running C Programs on Files:

- **C-Program to demonstrate function fread()/fscanf()**
- **C-Program to demonstrate function fwrite()/fprintf()**

C-Program to demonstrate function fread()/fscanf()

```
#include <stdio.h>

typedef struct {
    char name[50];
    int age;
} Person;

int main() {
    FILE *file;
    Person person;

    // Open a file in binary write mode
    file = fopen("data.bin", "wb");

    if (file == NULL) {
```

```
    printf("Could not open the file for writing.\n");
    return 1;
}

// Write data using fwrite()
Person persons[] = {{"Alice", 30}, {"Bob", 25}, {"Charlie", 35}};
fwrite(persons, sizeof(Person), 3, file);

// Close the file
fclose(file);

// Open the file in binary read mode
file = fopen("data.bin", "rb");

if (file == NULL) {
    printf("Could not open the file for reading.\n");
    return 1;
}

// Read data using fread()
printf("Reading data using fread():\n");
fread(&person, sizeof(Person), 1, file);
printf("Name: %s, Age: %d\n", person.name, person.age);

// Close the file
fclose(file);

// Open the file in text read mode
file = fopen("data.txt", "r");

if (file == NULL) {
    printf("Could not open the file for reading.\n");
    return 1;
}

// Read data using fscanf()
printf("\nReading data using fscanf():\n");
while (fscanf(file, "%s %d", person.name, &person.age) != EOF) {
    printf("Name: %s, Age: %d\n", person.name, person.age);
}

// Close the file
fclose(file);

return 0;
}
```

Output:

Reading data using fread():

Name: Alice, Age: 30

Could not open the file for reading.

C-Program to demonstrate function fwrite()/fprintf()

```
#include <stdio.h>
```

```
typedef struct {
    char name[50];
    int age;
} Person;
```

```
int main() {
    FILE *binaryFile, *textFile;
    Person persons[] = {"Alice", 30}, {"Bob", 25}, {"Charlie", 35};;
```

```
    // Open a binary file for writing
    binaryFile = fopen("data.bin", "wb");
```

```
    if (binaryFile == NULL) {
        printf("Could not open the binary file for writing.\n");
        return 1;
    }
```

```
    // Write data to the binary file using fwrite()
    fwrite(persons, sizeof(Person), 3, binaryFile);
```

```
    // Close the binary file
    fclose(binaryFile);
```

```
    // Open a text file for writing
    textFile = fopen("data.txt", "w");
```

```
    if (textFile == NULL) {
        printf("Could not open the text file for writing.\n");
        return 1;
    }
```

```
    // Write data to the text file using fprintf()
    for (int i = 0; i < 3; i++) {
        fprintf(textFile, "Name: %s, Age: %d\n", persons[i].name, persons[i].age);
    }
```

```
// Close the text file
fclose(textFile);

printf("Data written to files successfully.\n");

return 0;
}
```

Output:
Data written to files successfully.